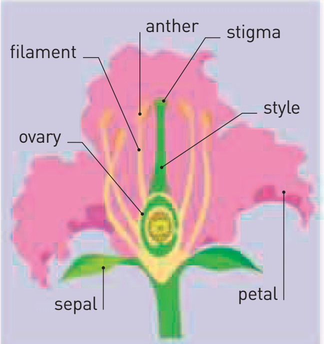
1831–32

British botanist Robert Brown was the first to observe and appreciate the significance of a plant cell's nucleus, orange in this modern colored scanning electron microscope image.



PLANT REPRODUCTION

Flowering plants reproduce sexually, and have both male (a stamen, made up of an anther and filament) and female parts (a pistil, made up of a stigma, style, and ovary). Fertilization begins when pollen from an anther lands on the stigma. A tube grows down the style to the ovary to deliver the male sex cells to the ovule (female sex cell).

regarded as the first true maps of the planet.

Italian microscopist and astronomer **Giovanni Battista Amici** (1786–1863) also studied flowers. He first noticed the **pollen tube**—a single cell tube that transports male sex cells to the plant's ovule—in 1824. In 1830, Amici used a microscope to observe the process whereby the pollen tube is formed.

THE STUDY OF ELECTRICITY AND MAGNETISM was proceeding at such a speed in the early 1830s that there were often disputes over who had made a particular discovery first. In 1831, British scientist **Michael Faraday** and his American counterpart **Joseph Henry** independently found that moving a magnetic field near a wire induced an electric current, thereby discovering the principle of **electromagnetic induction**. In time, this led to the development of machines that could generate large quantities of electricity, enabling advances such as electric lighting.

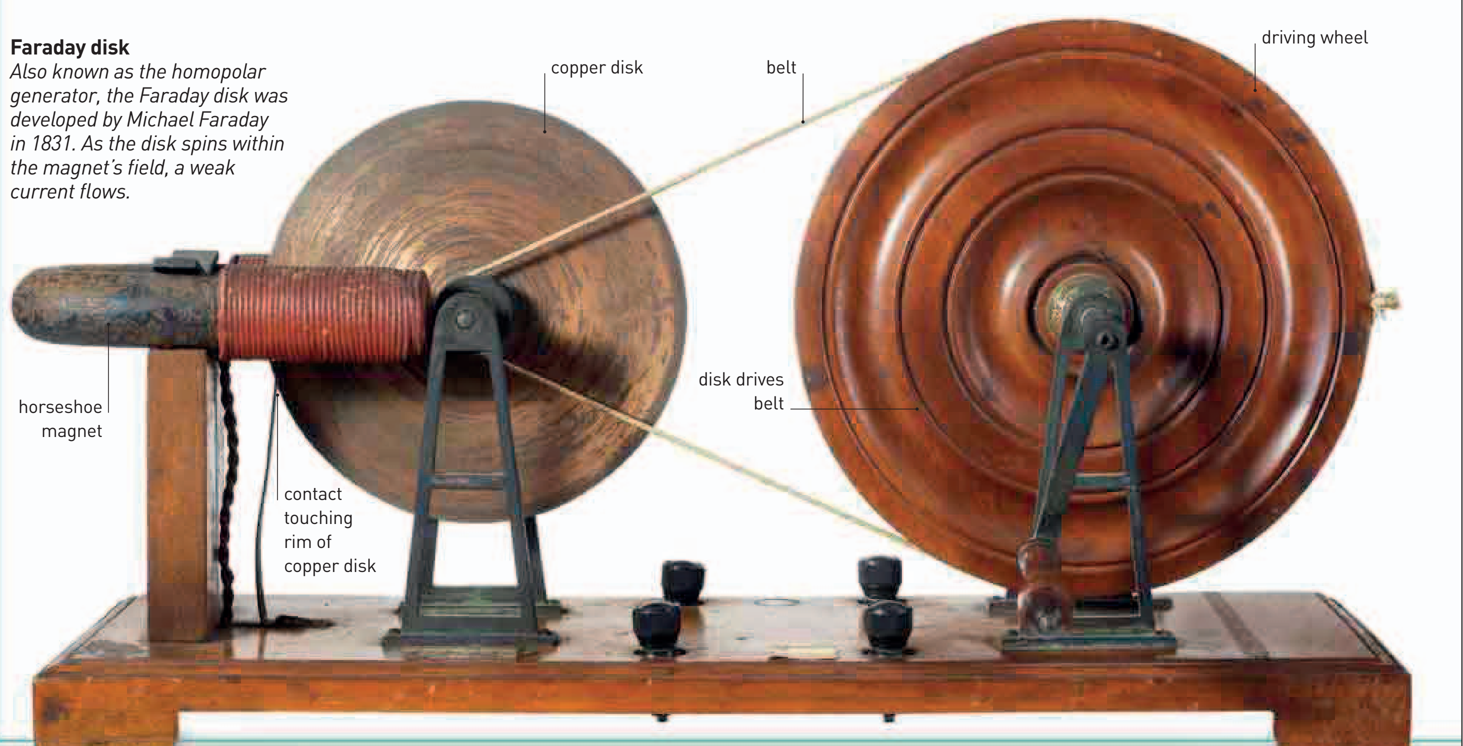
Less controversially, German mineralogist **Franz Ernst Neumann** (1798–1895) extended Pierre Dulong and Alexis Petit's discovery of the **inverse relationship of specific heat to the atomic weight of elements** (see 1819) to include molecules. Building up the picture of the relationship between atoms and molecules and the energy they carry, Neumann showed that the molecular heat of a compound is equal to the sum of the atomic heat of its constituents. This came to be known as **Neumann's Law**.

Swedish chemist **Jöns Jacob Berzelius** had already published a list of atomic weights of 43 elements in 1825. In 1831, he **introduced the term isomer** for different compounds with the same chemical composition.

In the same year, British botanist **Robert Brown** used the word **nucleus for the first time in biology**, to describe the central globule he saw through a microscope when he was observing the cells of orchids. Other scientists had seen the nucleus before, but Brown linked it to reproduction. In Germany, astronomer **Heinrich Schwabe** (1789–1875) made **the first detailed drawing of Jupiter's Great Red Spot** (see 1662–64).

In 1832, French instrument maker **Hippolyte Pixii** (1808–35) invented a magnet-electric machine, which was the **first direct current generator**, and British physician **Thomas Hodgkin** (1798–1866) described **Hodgkin's Lymphoma** for the first time.

Faraday disk
Also known as the homopolar generator, the Faraday disk was developed by Michael Faraday in 1831. As the disk spins within the magnet's field, a weak current flows.



Italian scientist Macedonio Melloni invents the **thermocouple**, a device to measure temperature

December Joseph Henry creates a **powerful electromagnet**

Foundation of **British Association for the Advancement of Science**

1831 Michael Faraday and Joseph Henry independently discover the phenomenon of **electromagnetic induction**

1831 Robert Brown discovers the cell **nucleus**

1831 Jöns Jacob Berzelius coins the term **isomer**

1831 Franz Ernst Neumann **extends the Dulong–Petit law** to include molecules and compounds

1831 Heinrich Schwabe draws **Jupiter's Great Red Spot**

December 27, 1831 British naturalist **Charles Darwin** embarks on his voyage in HMS ***Beagle***

1832 British physician Thomas Hodgkin describes **Hodgkin's lymphoma**

1832 French instrument maker **Hippolyte Pixii** invents one of the first machines to generate electricity